

Making a Piecewise Model Continuous

Directions:

- Each model below is built in pieces. A model is continuous at a junction $x = c$ when the left-hand limit, the right-hand limit, and the value $f(c)$ are all the same number.
- So set the two one-sided limits equal at the junction and solve for the unknown constant. Write the equation before you solve it.
- Units are given in every stem. Carry them through, and check that your constant has a sensible size for the quantity it belongs to.

1. An oven's temperature, in degrees Celsius, t minutes after it is switched on is modelled by

$$T(t) = \begin{cases} 20t + a, & 0 \leq t < 5 \\ 150 - 2t, & t \geq 5 \end{cases}$$

Find the constant a that makes T continuous at $t = 5$ minutes.

Answer: _____ degrees

2. A tank holds $V(x)$ litres of water x minutes after filling begins:

$$V(x) = \begin{cases} 4x + 6, & 0 \leq x < 6 \\ kx, & x \geq 6 \end{cases}$$

Find the constant k that makes V continuous at $x = 6$ minutes.

Answer: _____

3. A sensor reports $r(x) = \frac{x^2 - 25}{x - 5}$ for every pressure $x \neq 5$ bar, and its reading at $x = 5$ is a stored constant c . Choose c so that r is continuous at $x = 5$ bar.

Answer: _____

4. A drone's height above the launch pad, in metres, t seconds after launch is

$$h(t) = \begin{cases} t^2 + a, & 0 \leq t < 3 \\ 4t + 3, & t \geq 3 \end{cases}$$

Find the constant a that makes h continuous at $t = 3$ seconds.

Answer: _____ m

5. A calibration curve returns $E(x) = \frac{\sqrt{x+7}-3}{x-2}$ for every input $x \neq 2$, and the value stored at $x = 2$ is a constant c . Choose c so that E is continuous at $x = 2$. Give the exact value, not a decimal.

Answer: _____

- 6.** A three-stage conveyor speed, in metres per second, at position x metres along the belt is

$$f(x) = \begin{cases} x^2, & x < 2 \\ ax + b, & 2 \leq x < 5 \\ 3x + 1, & x \geq 5 \end{cases}$$

Both junctions must be continuous. Write the equation each junction gives, then solve the pair.

$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

- 7.** A phone plan charges $C(g)$ dollars for g gigabytes of data:

$$C(g) = \begin{cases} 30 + 12g, & 0 \leq g < 4 \\ a + 7g, & g \geq 4 \end{cases}$$

- (a) Find the constant a that makes C continuous at $g = 4$ gigabytes: $\underline{\hspace{2cm}}$
- (b) Suppose the company published $a = 45$ instead. Explain what that would do to the bill of a customer whose usage is very close to 4 gigabytes.

8. Challenge. A control system outputs

$$g(x) = \begin{cases} \frac{x^2 - 16}{x - 4}, & x < 4 \\ a, & x = 4 \\ bx, & x > 4 \end{cases}$$

where x is measured in seconds. Find the constants a and b that make g continuous at $x = 4$ seconds. All three conditions of continuity must hold.

$a =$ _____ $b =$ _____